

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary and Renal

Lecture: CPR 08

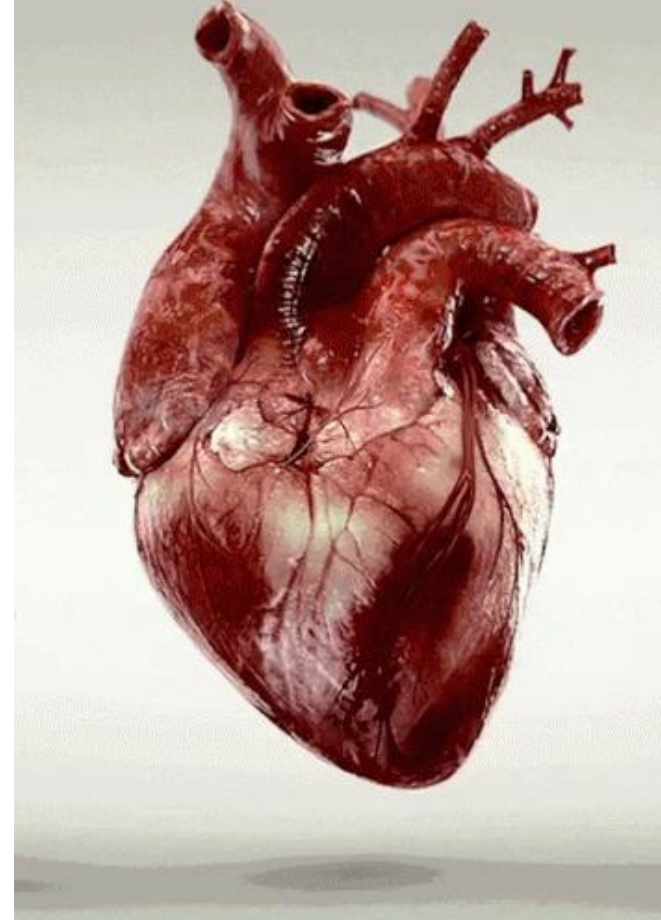
Determinants of Myocardial Performance

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Recommended readings

Chapter 13: Mechanics of the Cardiac Pump. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Cardiac cycle and pressure volume loops **must be reviewed** to be able to follow heart as a pump 1 and 2 concepts.

- Supplemental reading- really helpful:
- Mitral stenosis <https://www.cvphysiology.com/Heart%20Disease/HD009a>
- Mitral regurgitation <https://www.cvphysiology.com/Heart%20Disease/HD009c>
- Aortic regurgitation <https://www.cvphysiology.com/Heart%20Disease/HD009d>
- Aortic stenosis <https://www.cvphysiology.com/Heart%20Disease/HD009b>

Slides with blue background are the supplemental information associated with the previous slides and thus, might not be explained during the lecture time. However, please read the slides for clarification as they are equally important to know.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0103 Illustrate and explain the phases and events of the cardiac cycle using a diagram of a ventricular pressure loop.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0104 Differentiate between the calculations and definitions of stroke volume and stroke work. Compare the stroke volume and stroke work using a pressure-volume loop.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0105 Calculate ejection fraction from the values of end diastolic volume and end systolic volume/and or stroke volume.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0359 Describe the impact of changes in preload, afterload, and contractility in determining cardiac performance.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0344 Predict the change in ejection fraction that would result from a change in a) preload, b) afterload, and c) contractility.

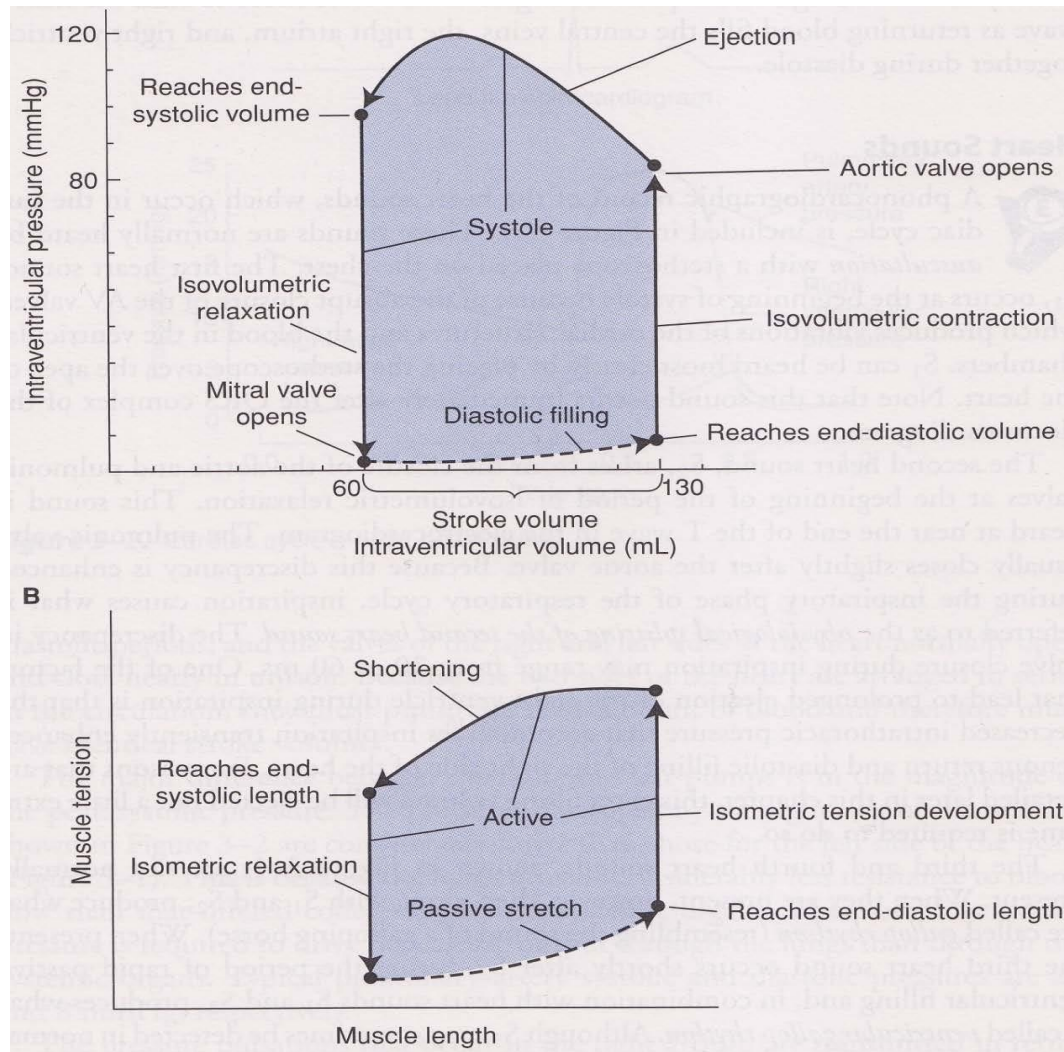
SOM.MK.II.BPM1.3.CPR.2.PHYS.0106 Describe the timing and causes of the four heart sounds.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0107 Describe the expected auscultation sounds and pressure changes (cardiac cycle, pressure volume loops) that define mitral stenosis, mitral insufficiency, aortic stenosis, and aortic insufficiency.

Lecture Outline

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- Pathological Heart sounds
 - S3, S4, (sometimes S1)
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 - Aortic Stenosis
 - Aortic Regurgitation
 - Mitral Stenosis
 - Mitral Regurgitation

Points to consider – the big picture



- Charts showing the relationship between the ventricular pressure – volume loop and the corresponding cardiac muscle length-tension cycle during one cardiac cycle.
- Relate points on both graphs to reference:
 - Preload
 - Afterload
 - Contractility
- Reflection of myocardial performance

Cardiovascular Physiology by Mohrman and Heller. Lange; McGraw Hill

Explanations for previous slide - relationship between the ventricular pressure – volume loop and the corresponding cardiac muscle length-tension cycle

- Cardiac muscle length-tension behaviour determines how the ventricles function.
- For each phase of the Pressure – Volume loop (one cardiac cycle) there is a corresponding phase of muscle length and tension changes.
- Diastolic filling = passive stretch of cardiac fibres
- End diastolic Volume sets the Preload.
- At onset of systole, ventricular muscles develop tension isometrically and intraventricular pressure rises.
- Once intraventricular pressure is greater than aortic pressure, aortic valves open, cardiac muscles now shorten (isotonic) and ventricular ejection begins. Aortic pressure is referred to as the afterload because it determines the tension that must be developed by cardiac muscle fibres before they can shorten.
- During ventricular ejection – cardiac muscle is generating active tension and shortening (i.e. afterloaded isotonic contraction).
- Stroke volume (amount of blood ejected) is determined by how far ventricular muscle cells are able to shorten during contraction. This depends on the length-tension relationship (i.e. the peak isometric tension curve) and the load against which they are shortening
- Once shortening stops, and the outlet valves closes, the cardiac cells relax isometrically and ventricular wall tension and intraventricular pressure fall during isovolumetric relaxation.

Myocardial performance – how to determine the work done by the heart

Basics: Two types of work done by heart to eject a volume of blood (SV) into arteries (aorta and pulmonary)

1. Stroke work: Contracting and pumping blood into aorta (and pulmonary artery)
2. Kinetic work: Accelerating the blood through the valves into arteries, ~ 1% of total work done (usually ignored)

If work is the force applied to an object times the distance the object moves

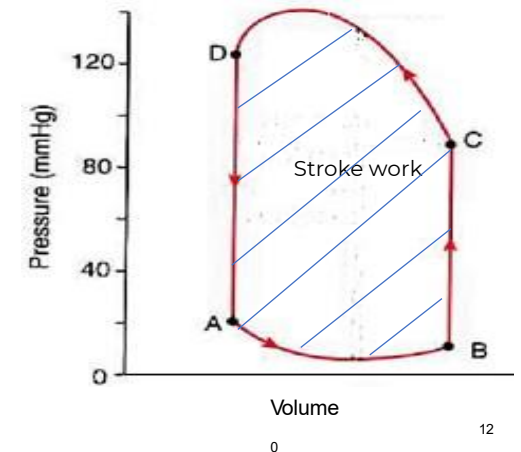
work done = force x distance, [force = pressure x area]

Stroke work = Pressure x Area x Distance, [volume = area x distance]

Stroke work = Pressure x Volume

Work done is the force (ventricular pressure) applied to a volume of blood (stroke volume) to eject it into the aorta.

Stroke Work = Stroke Volume x Mean Arterial Pressure

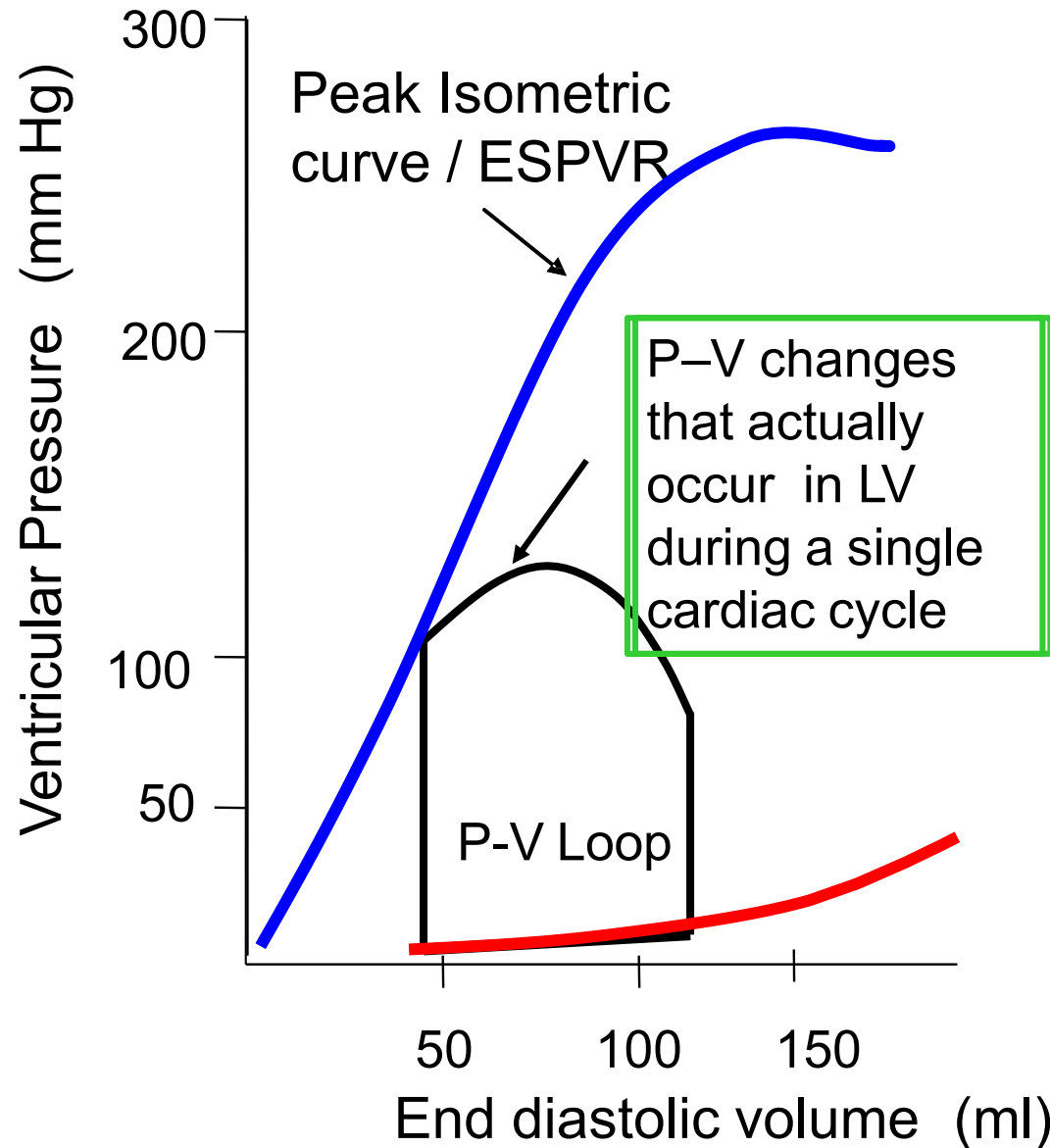


Stroke Work = Area of represented by the pressure – volume Loop

Factors that affect the work of the ventricles

- LV works 6x harder than RV
- (RV does not need to generate so much pressure to expel blood because pulmonary circulation is a low resistance circulation)
- Kinetic energy becomes significant when valves are stenosed.
- (Heart needs to work harder to push blood through narrowed valves).
- Large amount of work is converted to heat (“tension heat”)
- Heart consuming energy during isovolumetric contractions but no external work is being done. Energy expended ends up as heat.

Fitting in the Pressure-Volume Loop on the ventricular function curve



- Blue line is analogous to the peak isometric tension curve for isolated muscle.
- It limits how far shortening can occur.
- In the whole heart it sets the limit to how much blood can be ejected by a single contraction

Red line: Resting Tension =
Myocardial muscle fibre length
/End Diastolic Volume

Stroke volume = EDV- ESV
Ejection fraction = SV/ EDV

Explanation of previous slide:

In Cardiac Mechanics, the blue upper curve – is the peak isometric tension curve and the lower red curve is the resting tension in the myocardial fibres.

In the whole heart, the peak isometric tension curve is equivalent to the ventricular pressure developed in the ventricles and is called the End-Systolic Pressure-Volume Relationship (ESPVR).

The ESPVR defines the maximal pressure that can be generated at any given volume under a given inotropic state – equivalent to the Peak Isometric Tension curve

The resting tension is related to myocardial muscle fibre length which is related to the EDV.

Identifying changes in preload, afterload and contractility on the PV loop.

Mechanics - discussed the effects of changing preload, afterload and contractility on isolated myocardial fibres.

Determinants – discusses the effects of changing these parameters on the Pressure – Volume Loops.

Consider the immediate effects of changing one of the above parameters whilst keeping the other two constant.

Preload is changed by altering EDV.

EDV is altered by changing VR to the heart.

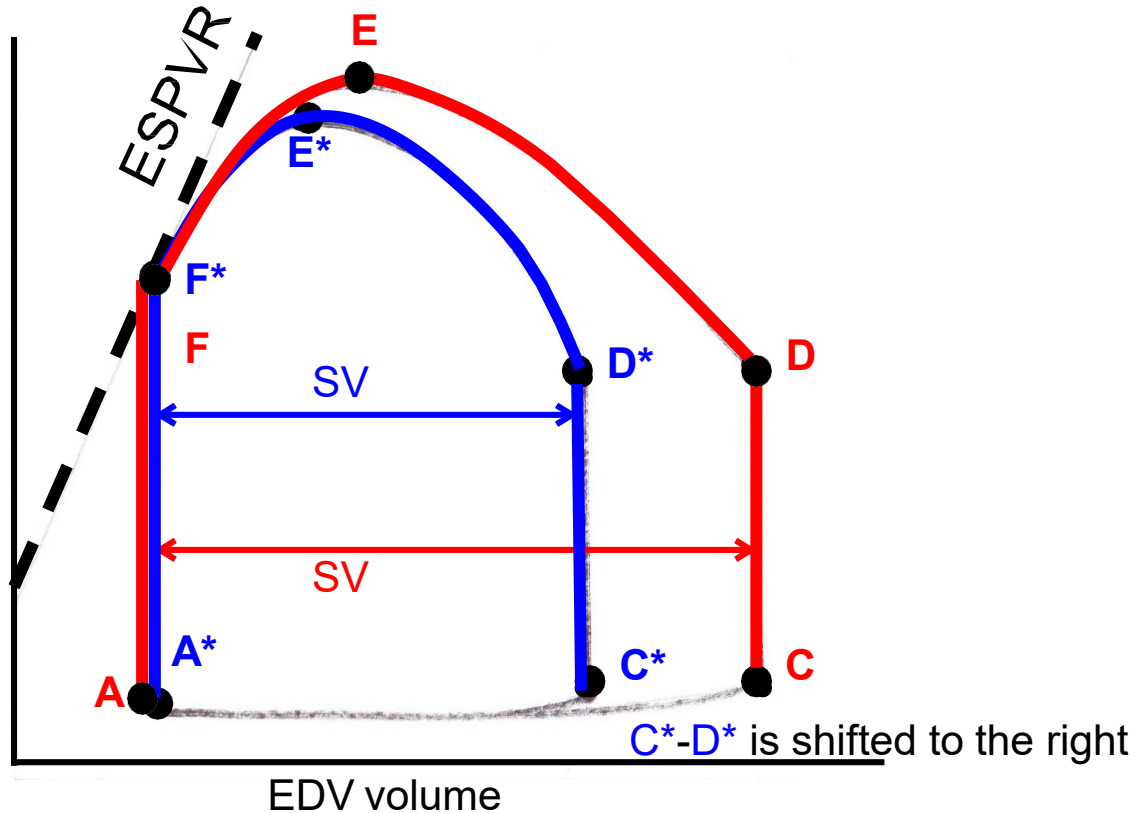
VR is altered by altering blood volume or by venoconstriction/venodilation or by slowing heart rate

Afterload is changed by altering aortic pressure or changing total peripheral resistance

Contractility is changed by altering sympathetic activity, or by giving drugs which alter inotropic state.

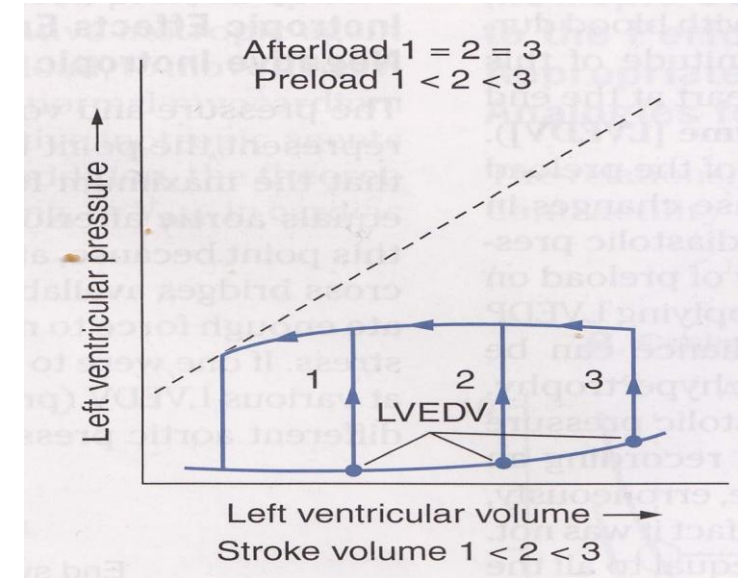
Effects of Increasing Preload

e.g. an acute response to an increase in VR,
Immediate effects *Due to Starling's mechanism*



$\uparrow VR = \uparrow EDV (= \uparrow \text{Preload})$

$\uparrow \text{Preload} \rightarrow \uparrow \text{Stroke Volume};$
Ejection fraction – maintained or slightly increased

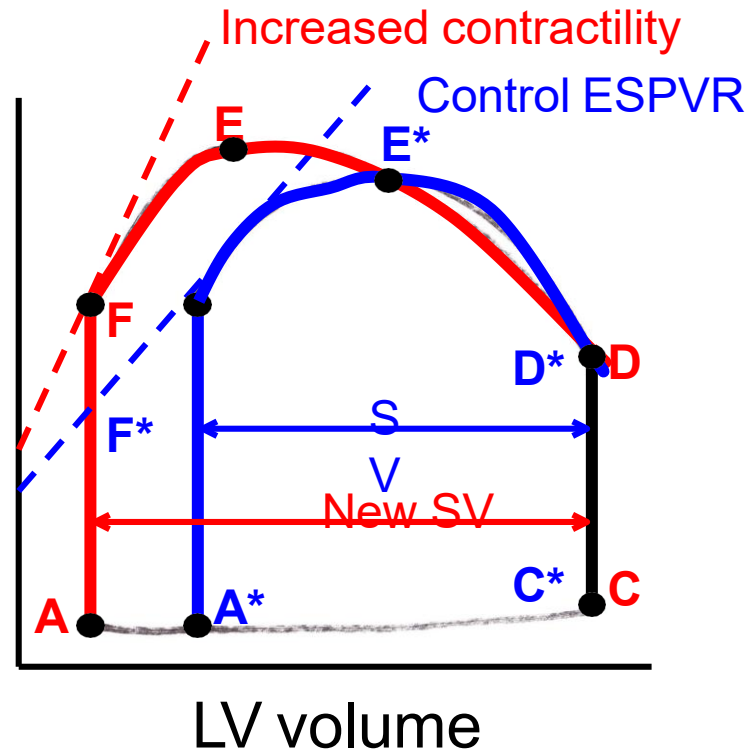


Cardiovascular Physiology by Mohrman and Heller. Lange; McGraw Hill

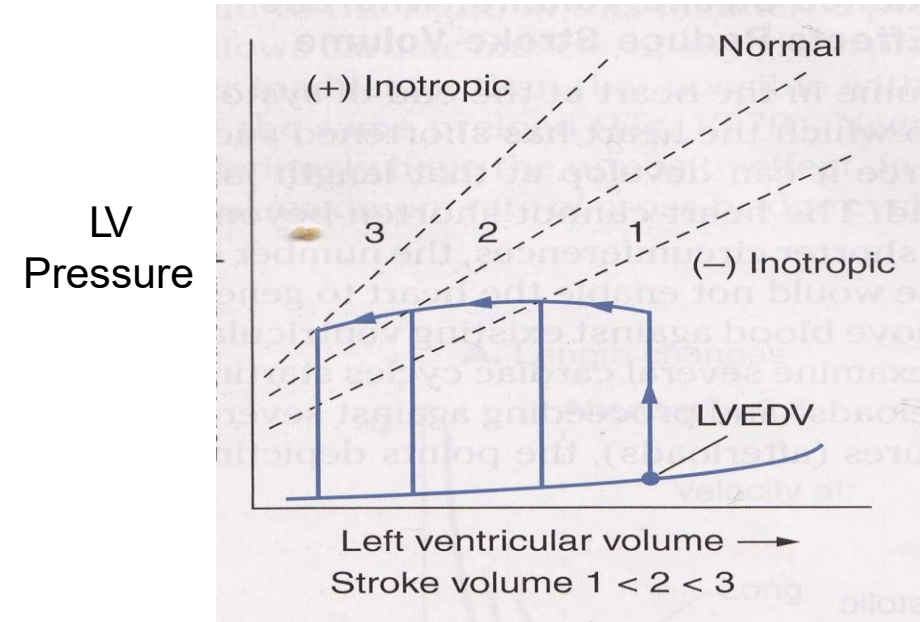
Effect of changes in LVEDV on stroke volume at constant arterial pressure (afterload) and inotropic state. (Frank-Starling relationship)

The greater the LVEDV the greater the SV

Effects of Increasing Contractility



Changes in contractility change the slope of the ESPVR; ESPVR becomes steeper

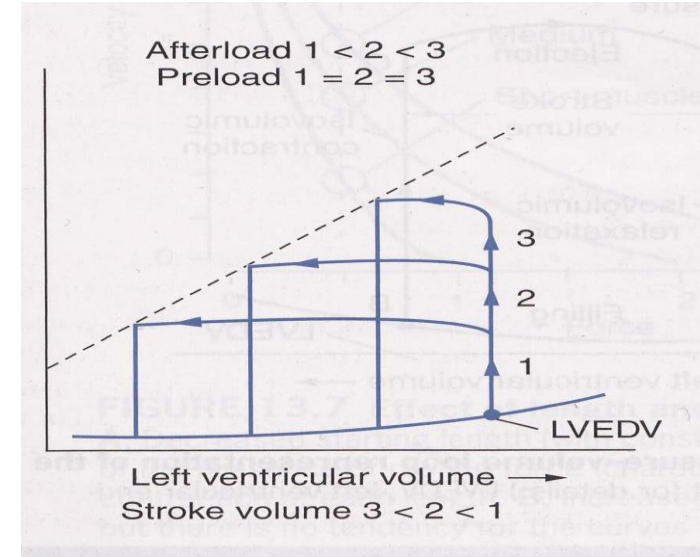
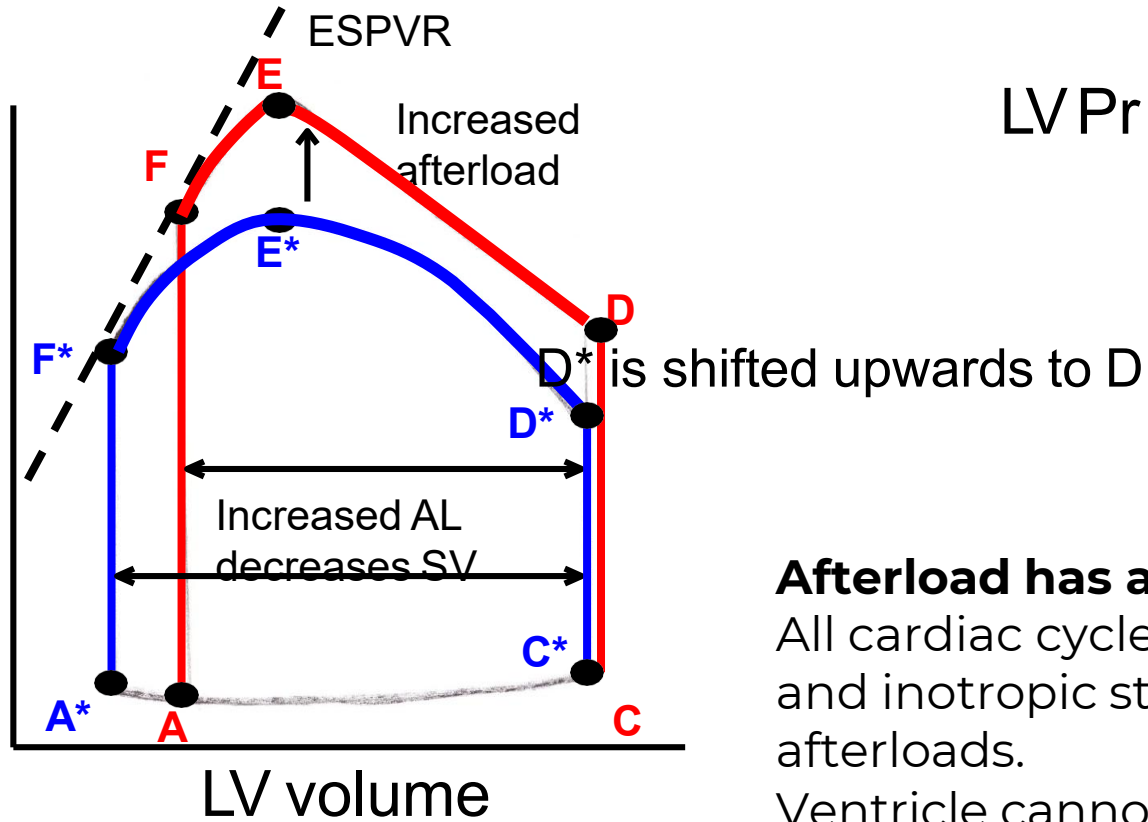


- Effect of inotropic state on stroke volume at a given LVEDV and arterial pressure (afterload).
- Increasing the inotropic state (contractility) of the heart increases SV from a given LVEDV
- SV is increased because muscle can shorten more.
- End Systolic Volume is reduced

↑ Contractility → ↑ Stroke Volume (stroke work) and Ejection Fraction

Effect of Increasing **Afterload**

e.g. Increased aortic pressure



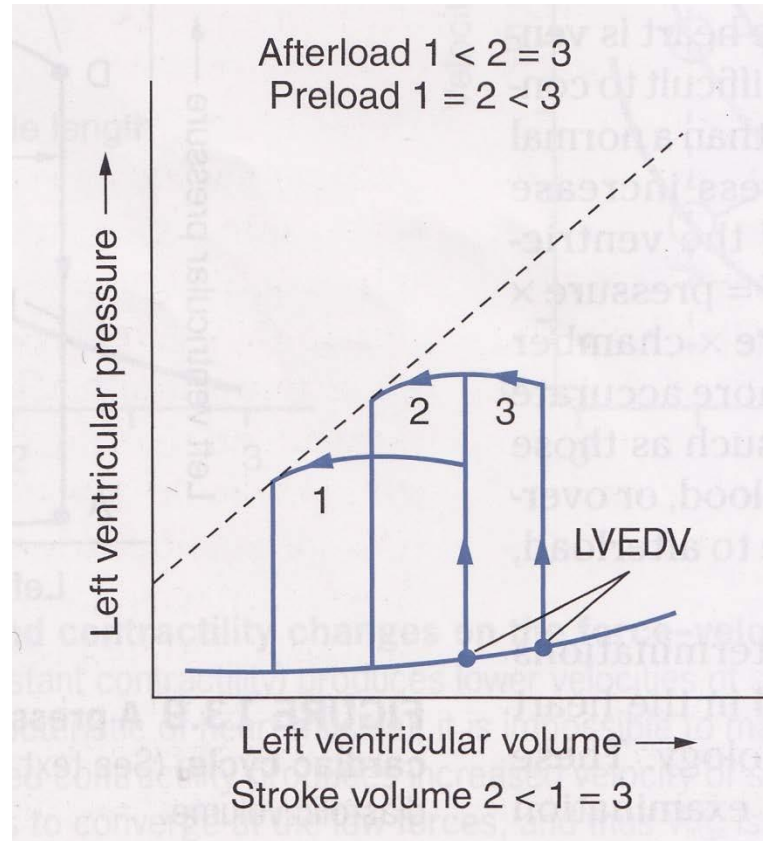
Afterload has a negative impact on SV.

All cardiac cycles here are starting from same LVEDV and inotropic state but proceed against 3 different afterloads.

Ventricle cannot open the aortic valve until the LV pressure reaches the higher aortic pressure. Because the ventricle has had to develop more isometric tension, its capacity to shorten is reduced, and the ventricle ejects less blood.

↑ Afterload → ↓ Stroke Volume and ↓ Ejection Fraction

Changes in LVEDV following an increase in afterload **and** during the next cycle



Loop 1 – contraction at normal LVEDV against a normal aortic pressure.

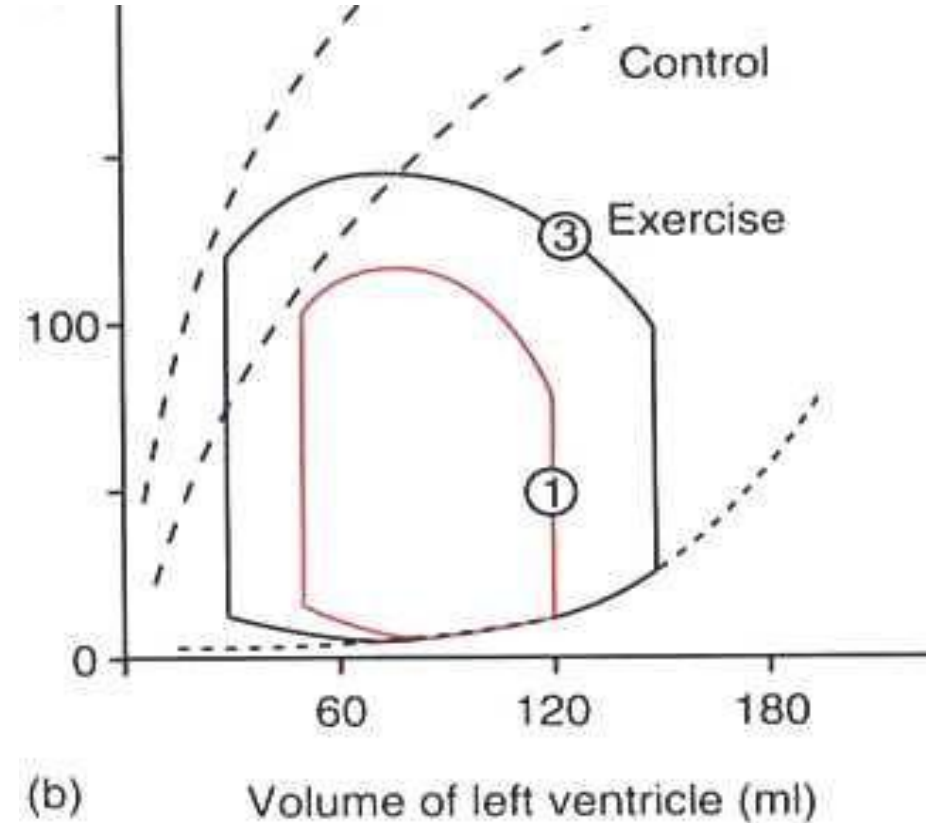
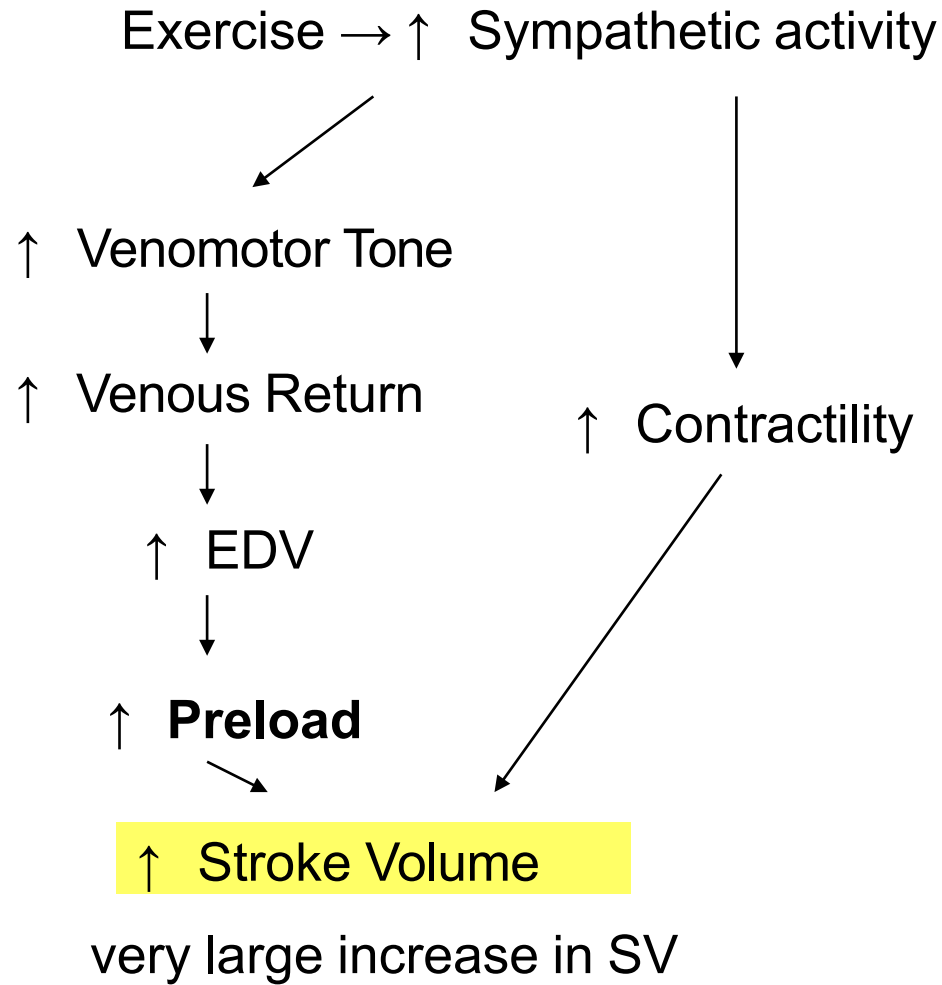
Loop 2 – first contraction against an $\uparrow$ aortic pr ($\uparrow$ afterload) from the same LVEDV – resulting in a $\downarrow$ SV.

Loop 3 - $\uparrow$ LVEDV increases because of increased end-systolic volume in last cycle. (ie. Preload has increased).

The increased preload increased resting tension, increasing stretch of myocardial fibre, SV is re-established after isotonic contraction.

This shows how Starling's Law of the heart can compensate for a fall in SV by increasing SV on the next cycle

Effects of Exercise on the Pressure – Volume Loop

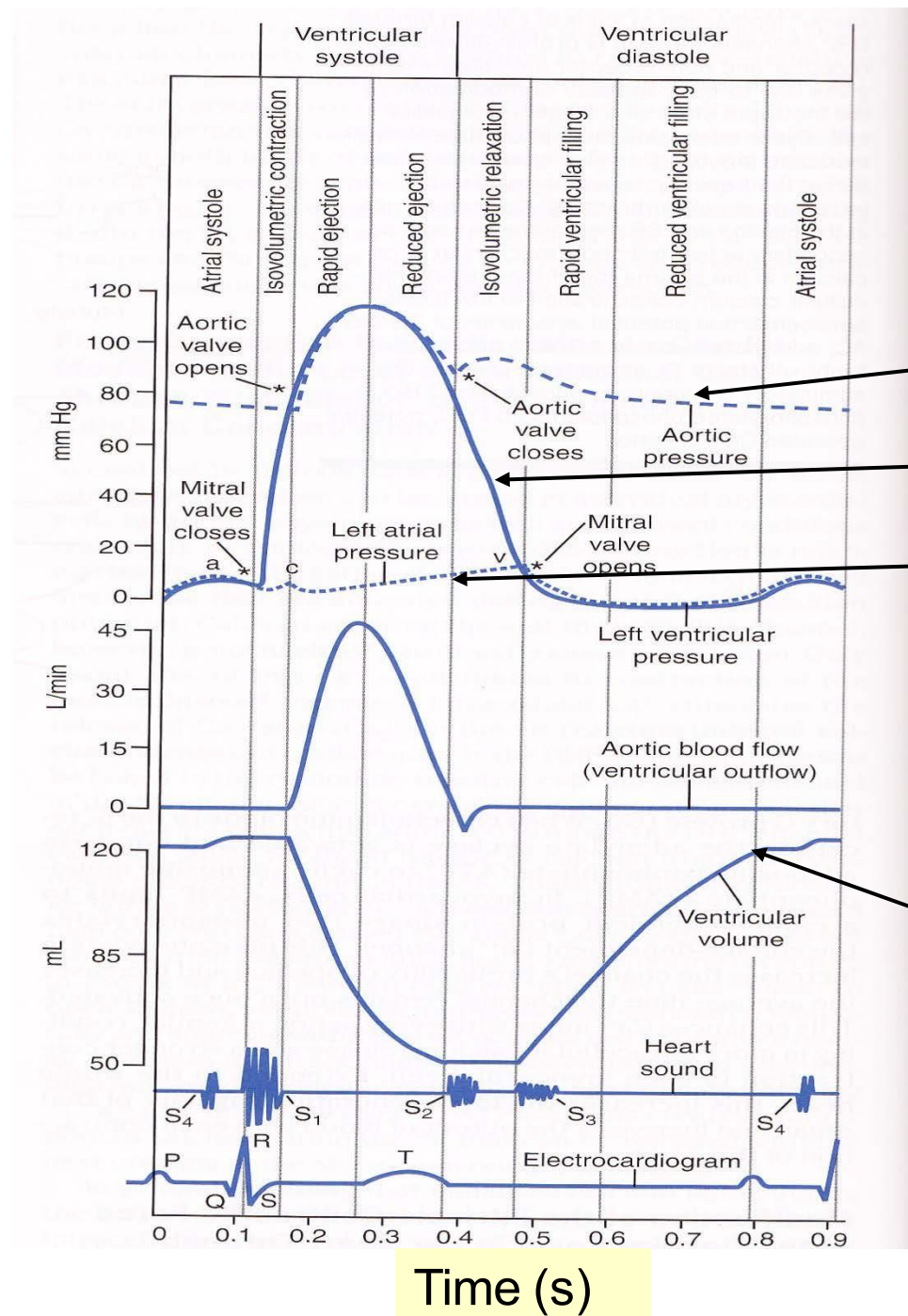


Lecture Outline

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The Cardiac Cycle

Wiggers Diagram



Aortic Pressure

LV Pressure

LA Pressure

Aortic blood flow

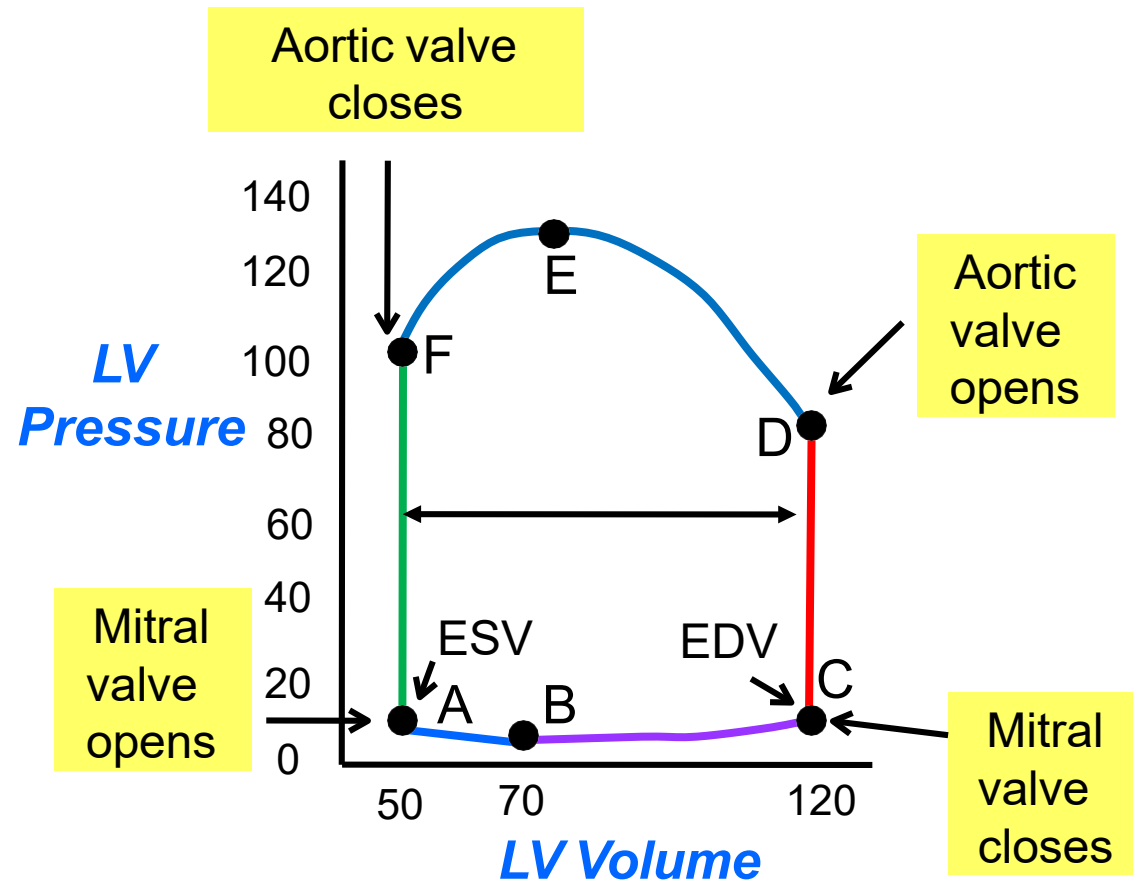
LV Volume

Heart Sounds

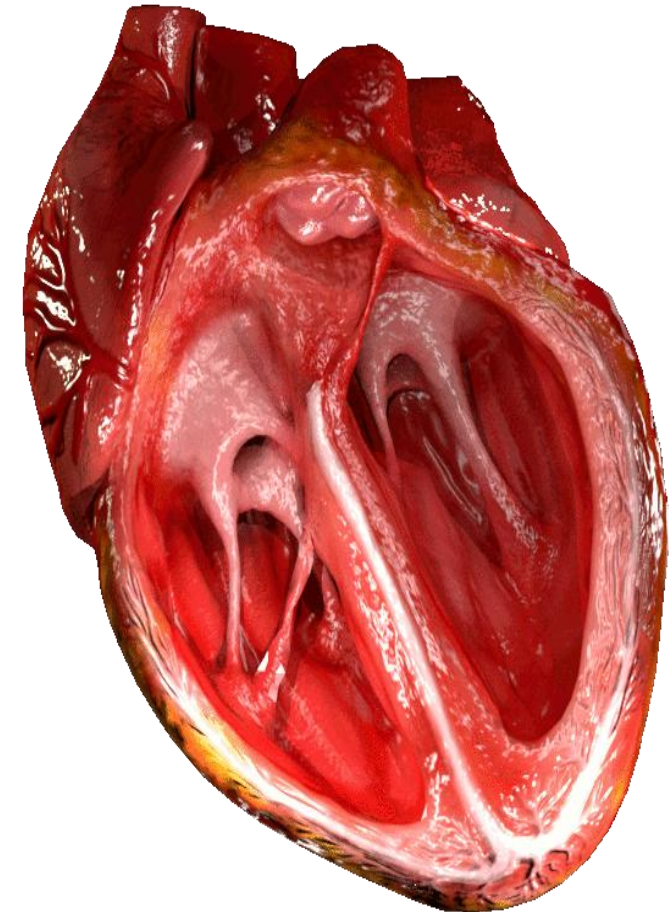
ECG

The heart sounds/valve closure on the PV loop

Pressure and Volume changes in the LV during one cardiac cycle



Which sound is lub and which sound is dub??

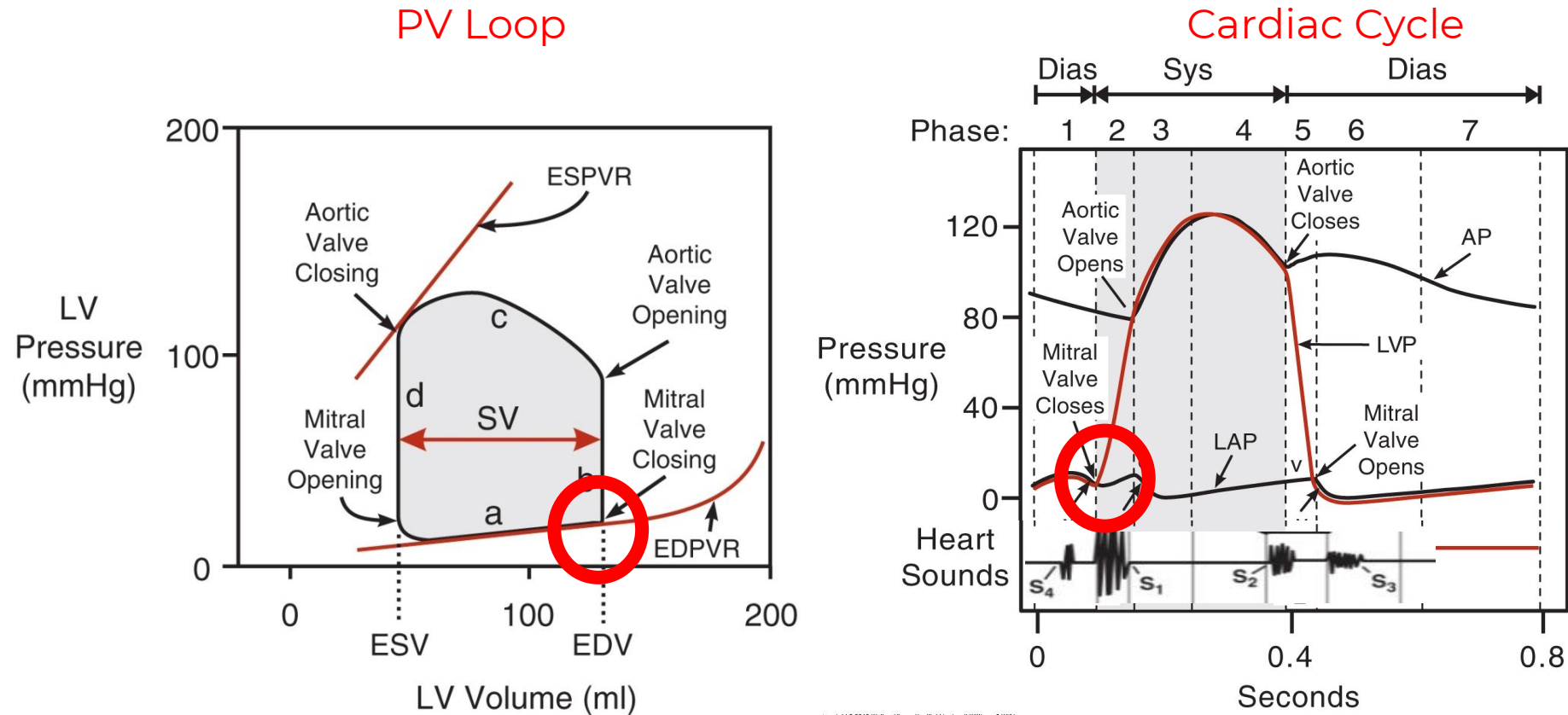


Powerful tool for analyzing ventricular function

https://upload.wikimedia.org/wikipedia/commons/b/bf/CG_heart_2.gif

When do you hear the heart sounds? – identify on PV loop and wigger

Lub = S1 : Closure of A-V valves (mitral and tricuspid)



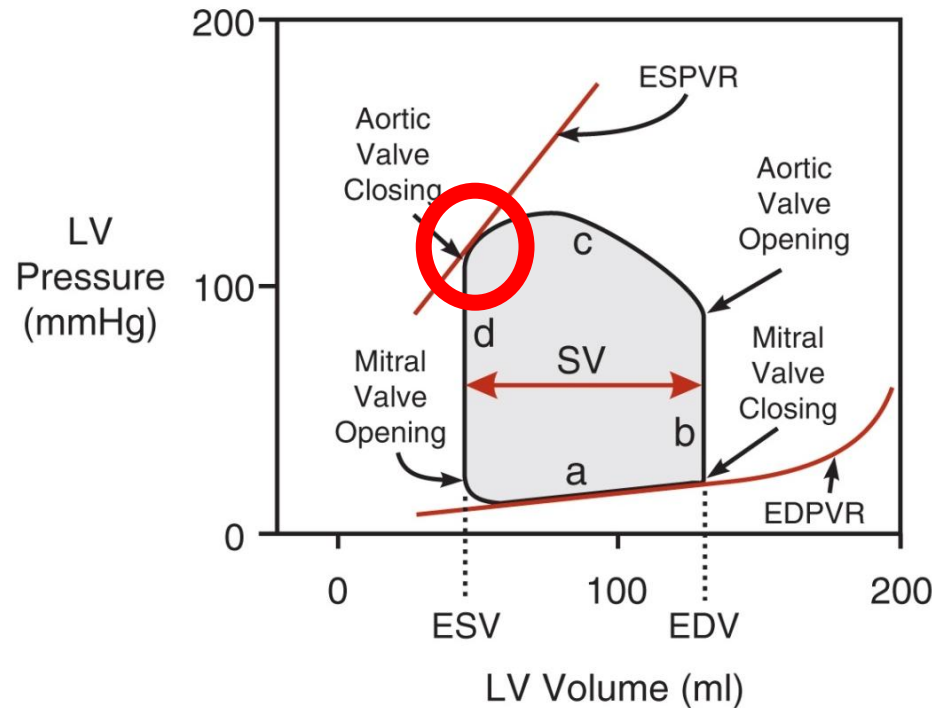
Large AV valves- low sound “lub”. A large drum (longer)

AV close when ventricle press > atrial press.

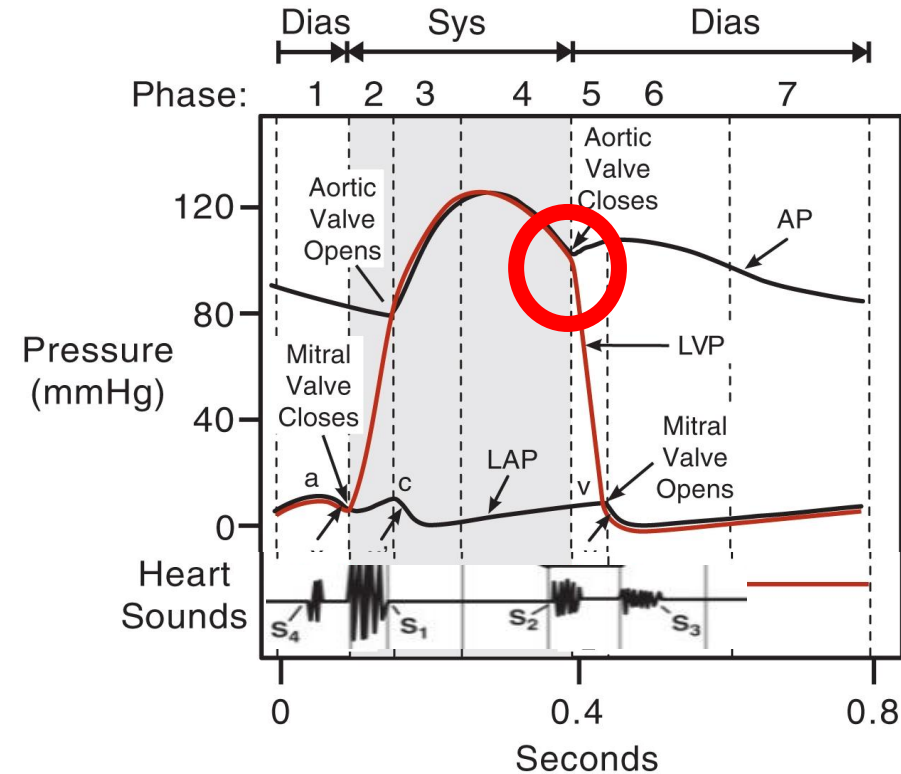
Valve closures occur at low pressures – atrial curves/ filling curves

When do you hear the heart sounds? – identify on PV loop and wigger

Dub = S2 : Closure of semilunar valves (aortic & pulmonary)



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Small semilunar valves- high pitch sound “dub”. A small drum
Semilunar valves close when ventricle press < AORTic press.
Valve closures occur at high pressures – aortic curves/ ejection
curves

Splitting patterns of the second heart sound.

Normal splitting *PHYSIOLOGIC*

Inspiration produce a drop in intrathoracic pressure , increasing venous return, increasing right ventricular filling, increasing right ventricular stroke volume, which produces a delayed closure of the pulmonic valve.

Wide splitting

Conditions that delay RV emptying:

- Pulmonic stenosis
- Right bundle branch block

Fixed splitting

Condition:

- Atrial septal defect: left-to-right shunt increases right atrium and right ventricle volumes, increasing flow through pulmonic valve which produce a delay in the pulmonic valve closure during both inspiration and expiration. Thus, the split is fixed (does not change with phases of breathing).

Paradoxical splitting

Normal order of valve closure is reversed (P2 occurs before A2). This is because aortic valve closure is delayed. Conditions:

- Severe Aortic stenosis
- Left bundle branch block

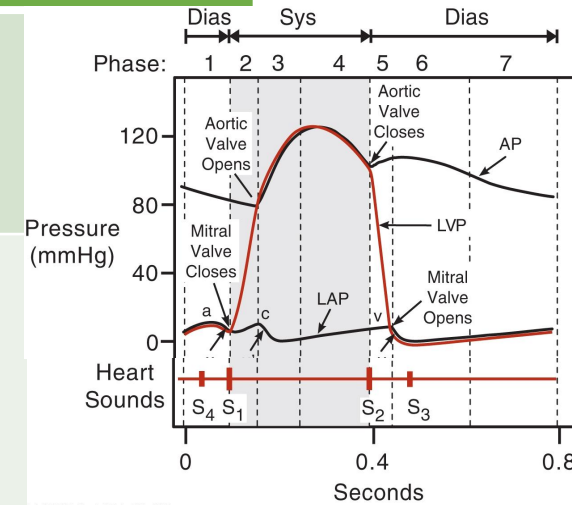
(In the above patients, during inspiration, P2 also closes later and moves closer to A2, paradoxically eliminating the split usually heard in expiration).

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Pathological Heart sounds: S1, S2, S3, S4

Heart sounds	Genesis & Timing	Audibility
S1 (MITI)	Occurs during mitral and tricuspid valve closure. <i>Conduction defects</i> Sound results from the vibrations of blood and the chordae tendineae in the ventricles.	Normally audible
S2 (A2P2)	Occurs during aortic and pulmonary valve closure. <i>normal</i> Sound results from the vibration of the valves and aortic wall. <i>Pathological if associated with conduction defects</i>	Normally audible
S3	Occurs in early diastole during rapid ventricular filling phase. Sound is originated by ventricular filling <i>recoil vibration</i>	It is not normally audible; may be audible in children <35 years
S4	Occurs in late diastole when atrial systole fills ventricle ("atrial kick"). Sound is originated by ventricular filling, <i>stiff ventricle with decreased compliance</i>	Not audible; if audible, indicates disease state >40 years



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Heart Murmurs

Abnormal sounds heard during Auscultation.

Caused by **FAULTY VALVES**

Types of Valve Abnormalities

- 1. Stenosis** Opened valve is **NARROWED**

Higher pressures needed to push blood through
- 2. Incompetence** Closed valve is **LEAKY** → **regurgitation**

A valve disorder can be termed e.g. aortic incompetence, aortic regurgitation or aortic insufficiency. It is the same!

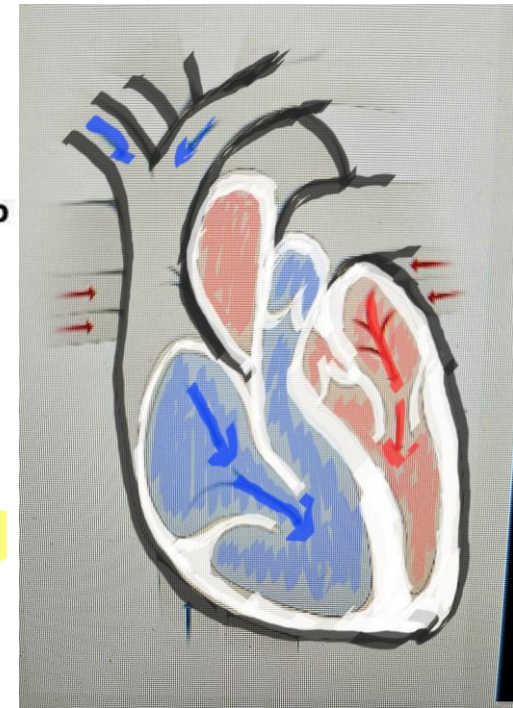
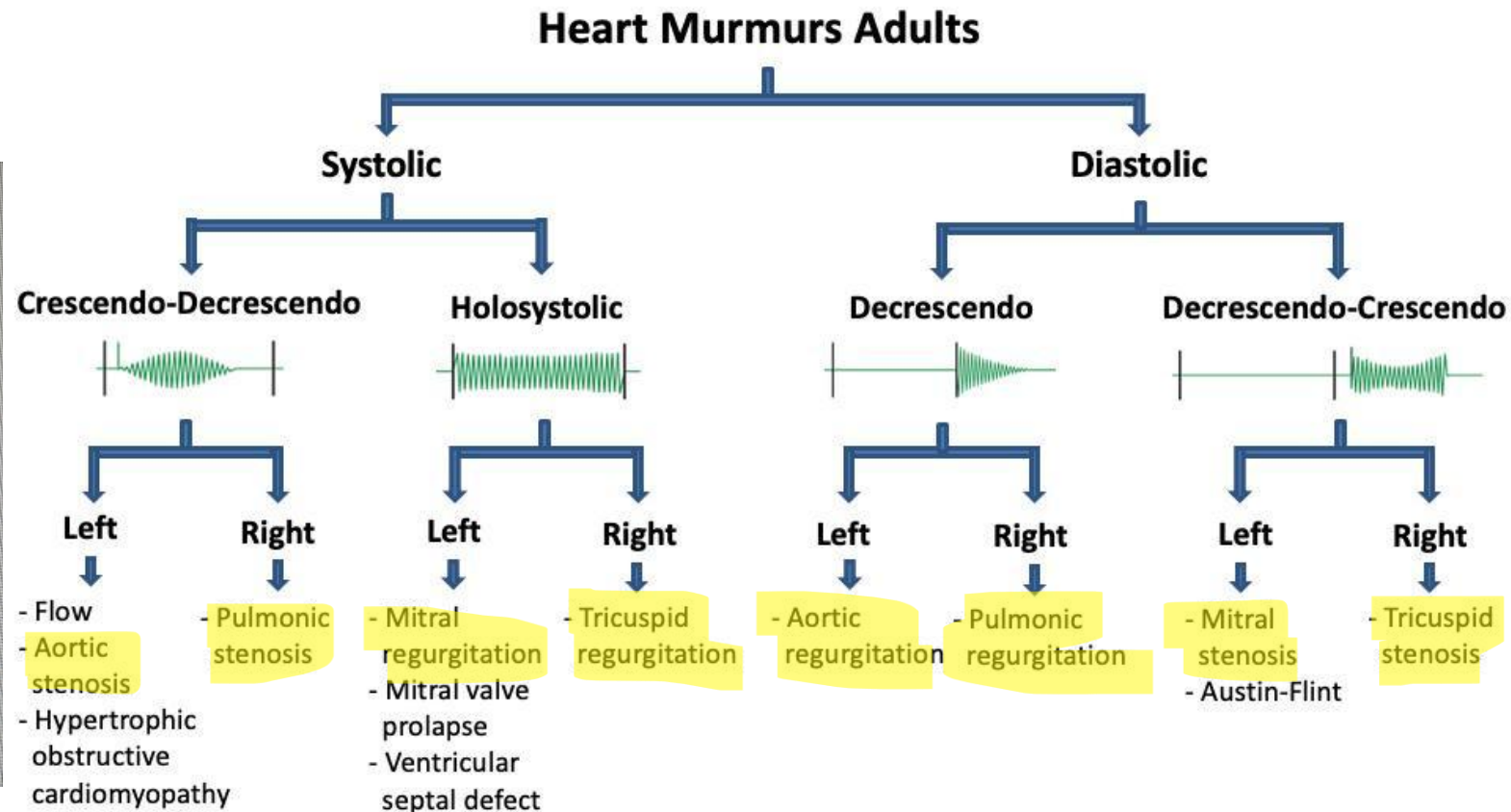
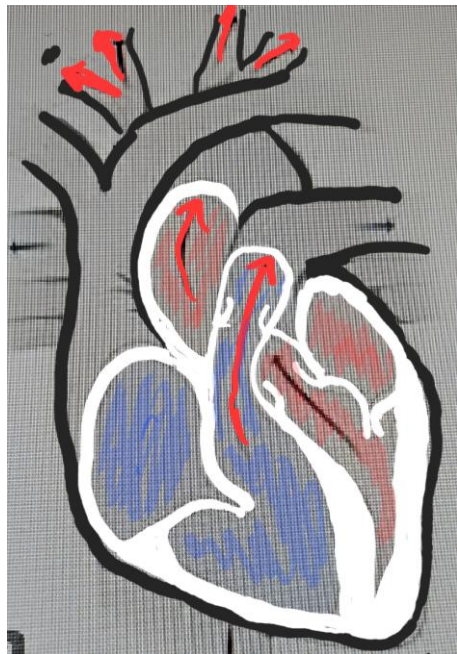
Faulty valves → **TURBULENT FLOW - NOISY**

Murmurs are described according to where they occur during cardiac cycle:

Each of 4 heart valves can be stenosed or incompetent/regurgitation.

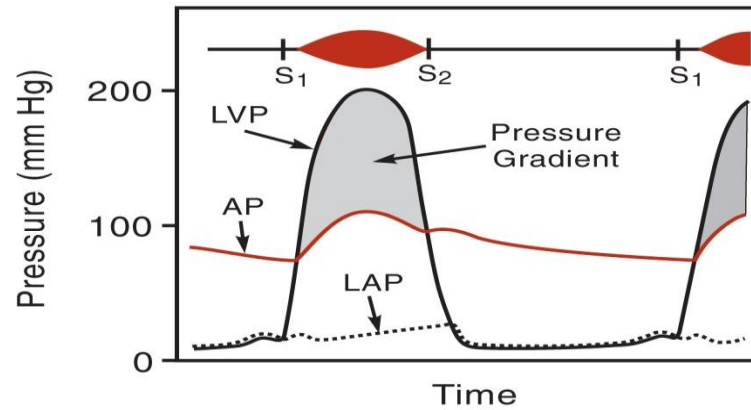
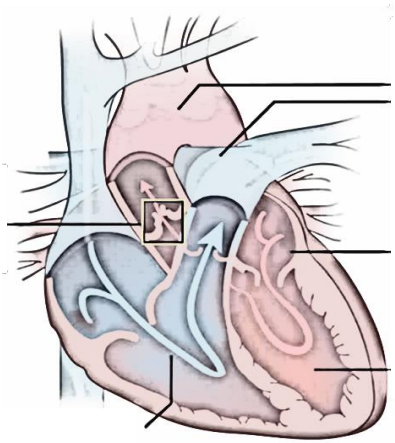
8 different pathologies

Systolic Murmur vs Diastolic Murmur

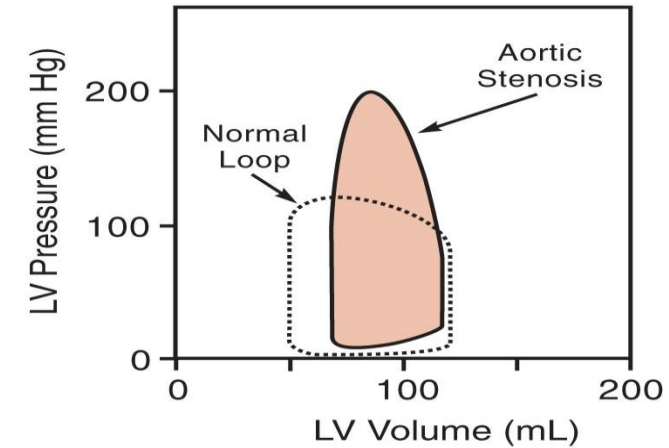


Systolic Murmur: Aortic Stenosis

Narrowing of aortic valve creates outflow resistance. Difficult to eject blood through aortic valve



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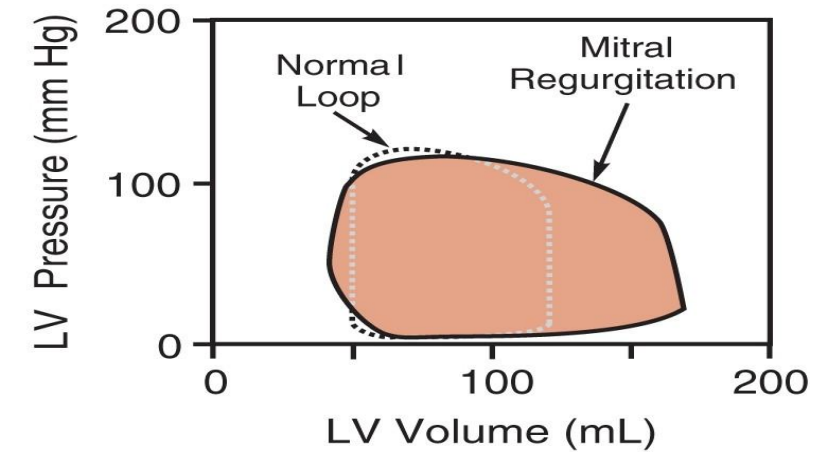
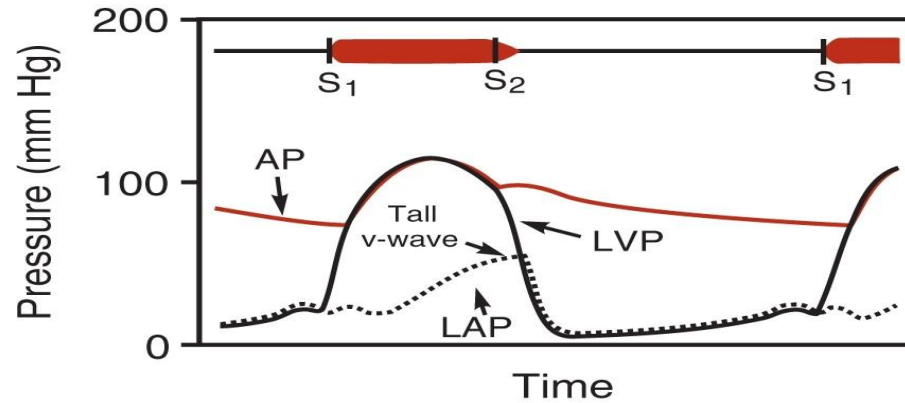
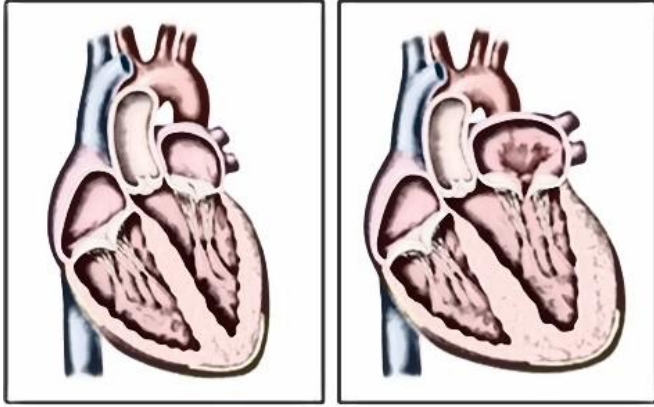
Murmur: crescendo-decrescendo systolic murmur, that does not extend beyond S2. *Murmur becomes more intense as flow increases through aortic valve and then lessens in intensity as ventricles start to relax.* S2 is ↓ in intensity;

Wiggers: Stenosis causes reduced outflow, which increases LV Pr. Low outflow but high velocity giving turbulence. During ejection LVPr exceeds aortic pressure. Large systolic pressure gradient (shaded) is present between LV and aorta. Note: Slow rising aortic pr. Left atrial pressure is elevated – backward pressure transmission

PV Loop: End-systolic volume is increased, little or no change in end-diastolic volume, SV is decreased. Compensatory Ventricular hypertrophy eventually reduces ventricular compliance which will increase LVEDPr and reduce LV filling. Increased Afterload

Systolic Murmur Mitral Incompetence (Regurgitation)

Ventricular systole causes blood to regurgitate back into LA resulting in murmur that extends throughout systole.



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Murmur: Pan systolic murmur (heard throughout systole)

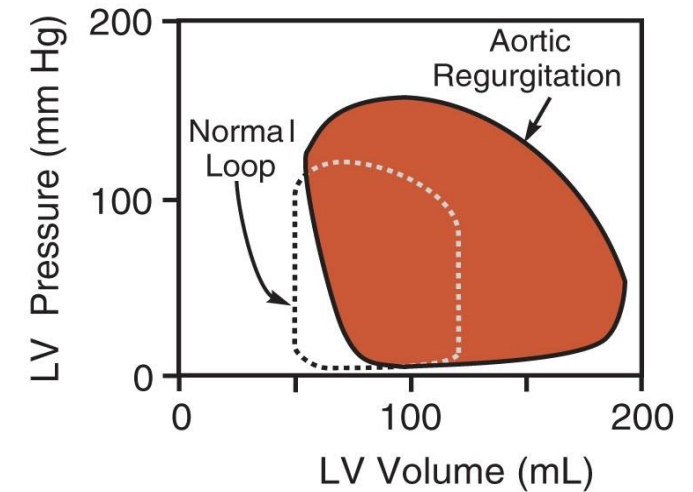
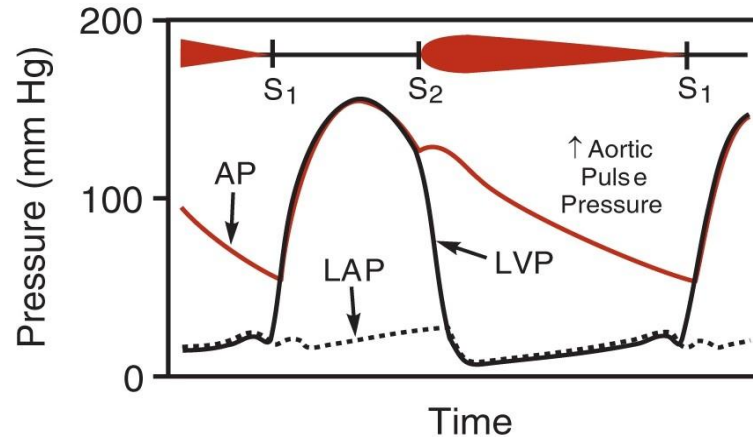
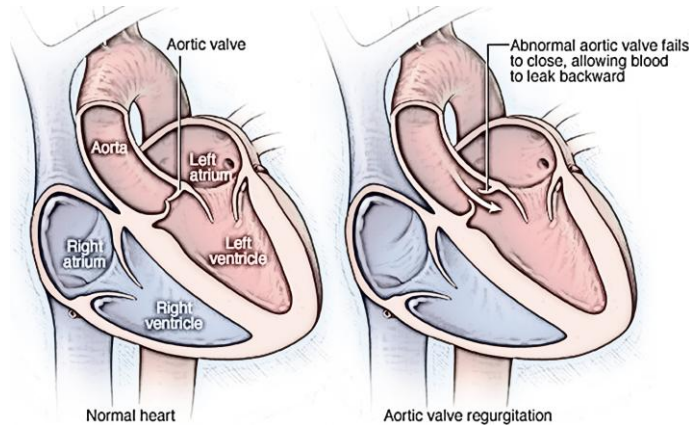
Wiggers: During ventricular contraction, the LV ejects blood back into the LA as well as into the aorta. Thereby increasing LA Pressure, particularly the v wave.

The aortic pr and LV Pr may fall in response to a reduction in the net volume of blood ejected into the aorta. With high volume regurgitation the forward cardiac output is reduced.

PV Loop: End diastolic volume is increased because increased LA pressure increases ventricular filling; SV is greatly enhanced. End systolic volume is reduced because of decreased outflow resistance; No true isovolumetric phases!

Diastolic Murmur Aortic Incompetence (Regurgitation):

Aortic valves fail to close properly LV pressure rises as blood regurgitates from aorta into LV.



Murmur: Decrescendo murmur. *In early diastole the pressure difference is greatest between the aorta and ventricle hence the louder the sound in early diastole.*

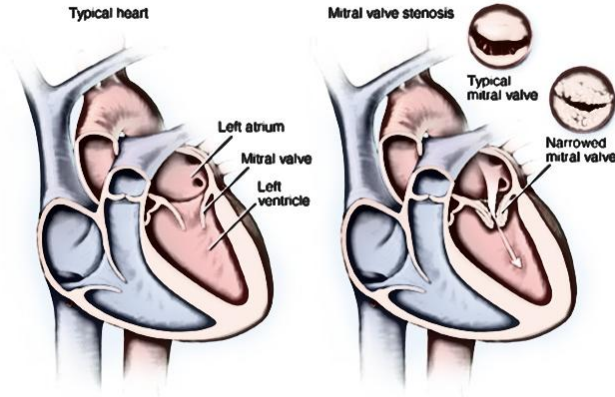
Wiggers: During diastole, ventricular relaxation, the leaking aortic valve allows blood to flow from aorta into the ventricle causing a rapid drop in aortic pressure due to blood volume leaving the aorta. This decreases diastolic pressure and increases aortic pulse pressure. Increased filling leads to an increase in peak ventricular and aortic pressure during ejection.

PV Loop: LV EDV and LV EDP are greatly increased, and LA pressure increases because of blood backing up into the atrium. There are no true isovolumetric phases.

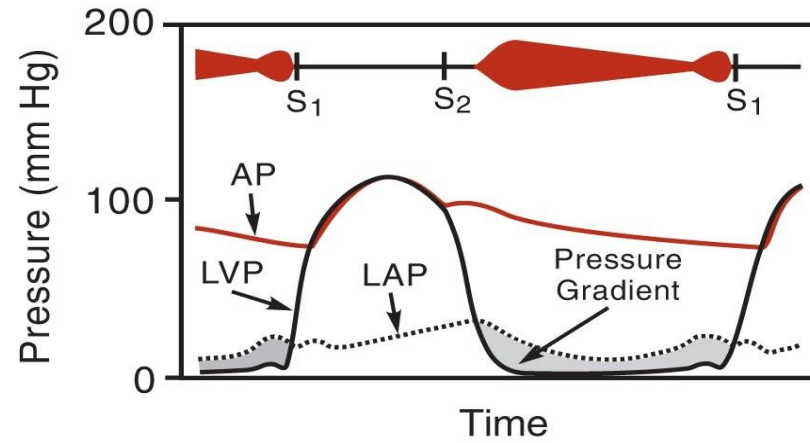
Increased LVEDV and filling leads to an increase in in ventricular stroke volume.

Diastolic Murmur **Mitral Stenosis:**

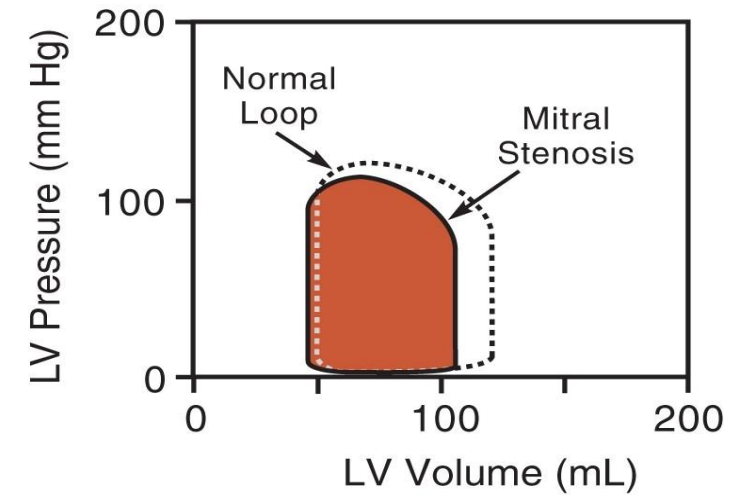
Mitral valves fail to open properly LA pressure rises as blood remains in LA during ventricular filling



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Murmur: Begins after S2 (closure of aortic/pulmonic valve), ventricles are relaxing, and is preceded by an opening snap (OS). *OS is a high pitched sound which may be due to sudden tensing of the stenotic leaflets on opening of the valve. Diastolic rumble. The interval between S2 and OS is inversely related to the severity of the MS.*

Wiggers: During ventricular filling, LA pressure exceeds LV pressure; Aortic pressure is reduced by severe mitral stenosis because of decreased cardiac output.

PV Loop: End-diastolic volume is reduced because of impaired ventricular filling, and end-systolic volume may be slightly reduced because of reduced afterload; therefore stroke volume is reduced.

Summary sheet Valve Abnormalities and Presenting Features

Problem	Murmur phase	Comments	Features to look for in Wiggers diagram	Features to look for in a P-V Loop
Mitral stenosis	Diastole	Opening Snap! Early diastolic with presystolic kick – without atrial fibrillation	LA Pr significantly elevated above LV Pr esp in diastole	Low LVEDV, low SV, low aortic diastolic pressure
Mitral Regurgitation	Systole	Holosystolic , high-pitched “blowing murmur” Loudest in the apex and radiates toward the axilla	Significantly elevated LA Pr during ventricular systole	No isovolumetric phases
Aortic Stenosis	Systole	Crescendo-decrescendo systolic ejection murmur	LV Pr significantly elevate above aortic pr during vent systole	High LVEDV, high peak systolic pr
Aortic Regurgitation	Diastole	High pitched “blowing” early diastolic decrescendo murmur.	Rapid drop of aortic pr in diastole, slower drop in LV pr during relaxation.	No isovolumetric phases